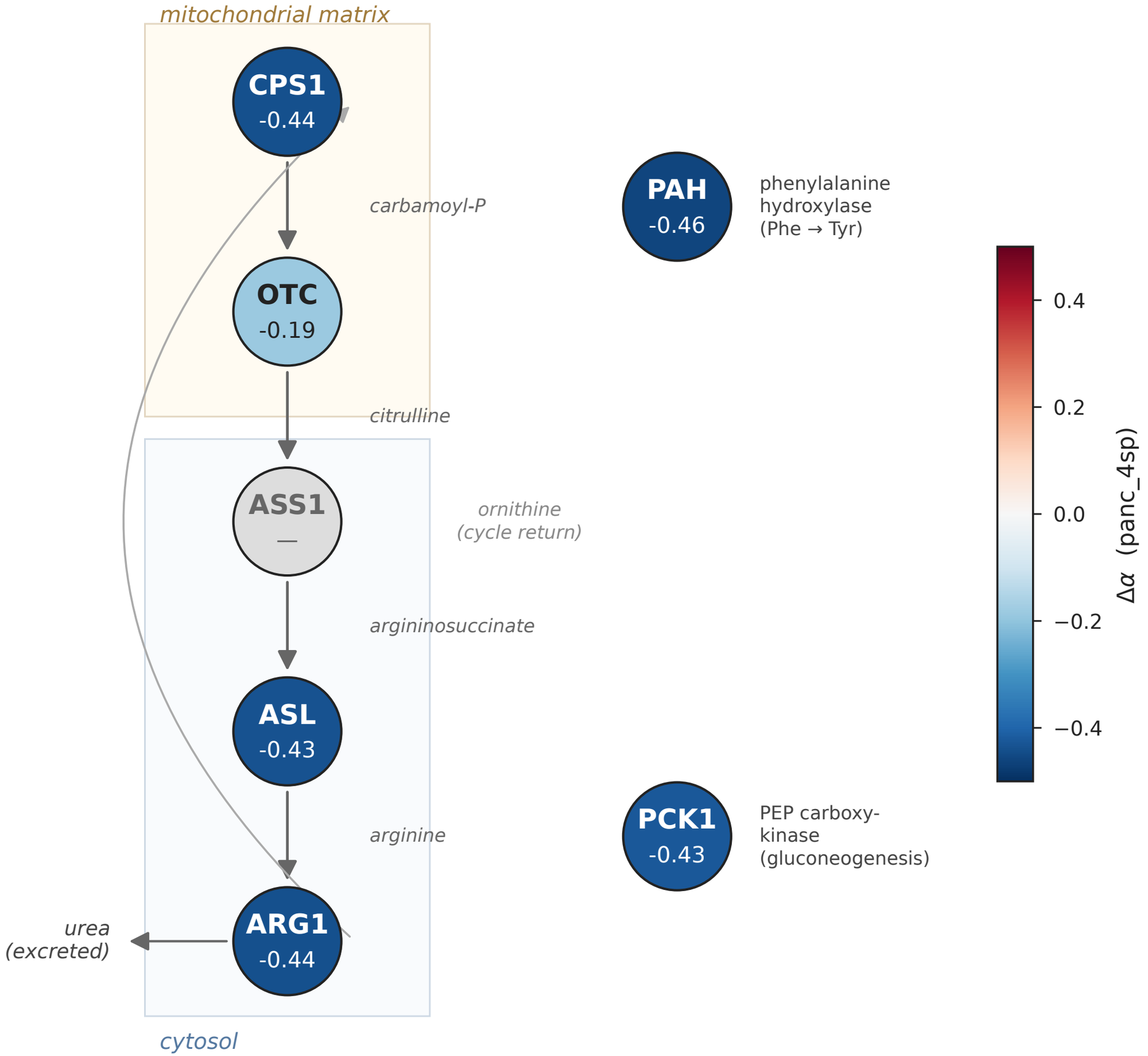
mitochondrial matrix
CPS1
-0.44
carbamoyl-P
OTC
-0.19
citrulline
ASS1
—
ornithine
(cycle return)
argininosuccinate
ASL
-0.43
arginine
ARG1
-0.44
urea
(excreted)
cytosol
PAH
-0.46
phenylalanine
hydroxylase
(Phe → Tyr)
PCK1
-0.43
PEP carboxy-
kinase
(gluconeogenesis)
0.4
0.2
0.0
−0.2
−0.4
Δα (panc_4sp)